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AI3403: Multi-Agent Systems (MAS) - Assignment #5

Rules for Submitting Assignment

- The assignment is worth 100 points. You must present the entire work leading to their solution and also attach the associated code.
- The assignment will be due by **23:59 on Friday, October 30, 2026** in the Google classroom. The deadline is strict and no extensions shall be awarded unless otherwise for unavoidable circumstances.
- Usage of AI tools like ChatGPT/Claude/Gemini is permitted and must be limited only to understand the concepts taught in the lecture. If you use them in the assignment, you should explicitly declare where and to what extent you have used it. Educate yourself enough to know and spot when the AI tool that you are using is giving you some nonsense!
- You should submit your assignment as **one** pdf document and the file should be named using roll number. For instance, if your roll number is **ai25btech11111**, then the pdf should be named as **ai25btech11111Assignment5.pdf**. Please also write your name in the first page of your assignment.
- Students are strongly encouraged to submit their work in a neatly typed LaTeX document with all supporting code attached to the same document. An example overleaf latex template can be found in this link: `https://www.overleaf.com/latex/templates/assignment-homework-solutions-template/nfjybpqbdrrn`. If typing is not possible, hand-written submissions will be considered as long as they are legibly written and scanned properly with all the codes attached into one pdf.
- Students can use any coding language of their preference such as Matlab, Python, Julia, C++, etc. All the associated codes for the assignment problems should be attached at the end of each respective problem in your assignment submission. You can also use Jupyter notebook, Google colab, etc to do your assignment and submit it as one pdf.
- You can use internet, any books, lecture notes, etc. Discussing with other students is perfectly fine and in fact is strongly encouraged. But plagiarism (copying solutions from others without putting your hard work) will never be tolerated.

Problem 1: Cooperative Navigation Problem (25 pts)
Consider a 5×5 grid with coordinates (x, y) where $x, y \in \{1, 2, 3, 4, 5\}$. Two agents (Agent 1 and Agent 2) move on the grid with actions $\mathcal{A}_i = \{\mathsf{Up}, \mathsf{Down}, \mathsf{Left}, \mathsf{Right}, \mathsf{Stay}\}$, for $i \in \{1, 2\}$. Actions that would move an agent outside the grid leave it in place (wall). Let the positions of both the agents at time t be $p_{1,t} = (x_{1,t}, y_{1,t})$ and $p_{2,t} = (x_{2,t}, y_{2,t})$. There are two landmarks g_1 and g_2 represented using grid cells. An episode ends when both landmarks are covered, meaning that at time t, there exists a one-to-one assignment of agents to landmarks such that each landmark has one agent on it (agents can swap roles). Episodes also terminate at a fixed horizon $H = 30$ if not yet successful. At each step, the team reward is

$$r_t = \begin{cases} +20 - 1, & \text{if both landmarks become covered at time } t, \\ -5, & \text{collision penalty if } p_{1,t} = p_{2,t}, \\ -1, & \text{otherwise,} \end{cases}$$

1. (5 points) Formulate the above environment as a MAMDP and specify the tuple details along with the termination condition.
2. (5 points) Assume each agent runs Independent Q-Learning (IQL) with a tabular $Q_i(s, a_i)$ and ε-greedy exploration. Write the IQL update equation for Q_i, $i = \{1, 2\}$ and explain why the learning problem becomes non-stationary from the viewpoint of an individual agent, even though the environment dynamics are fixed.
3. (5 points) Explain the key idea of Centralized Training with Decentralized Execution (CTDE) in MARL. Describe mathematically a centralized action-value function $Q(s, a_1, a_2)$ and how it can be used during training. Give two concrete reasons why CTDE helps compared to purely independent learning in cooperative tasks.
4. (10 points - Coding Task) Implement Independent Q-Learning (IQL) for the two-agent grid-world. Choose S be a random positive integer less than 10^4. Define $k = \text{mod}(S, 5) + 1$. Define the target location as $g_1 = (1, k)$, $g_2 = (5, 6 - k)$ and Let the initial positions of both the agents be $p_{1,0} = (1, 1), p_{2,0} = (5, 5)$. Use $\alpha = 0.2, \gamma = 0.95, \varepsilon_0 = 1.0, \varepsilon_{\min} = 0.05, \varepsilon \leftarrow \max(\varepsilon_{\min}, 0.995\,\varepsilon)$ per episode. Train for $N_{\text{ep}} = 2000$ episodes with horizon $H = 30$. Your submission must include:
   - A plot of the moving-average episode return (window 50).
   - A plot of the moving-average success rate (window 50), where success means both landmarks covered.
   - Print greedy (no exploration) trajectories for 3 evaluation episodes after training from fixed initial positions, showing both agents' step-by-step positions.

Problem 2: Distributed MPC Feasibility with Competing STL Specifications (25 pts)

Consider a swarm of four mobile robots moving in a two-dimensional plane. The robots are indexed by $\mathcal{V} = \{1, 2, 3, 4\}$. The communication topology is described by the cycle graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{E} = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$. For each robot $i \in \mathcal{V}$, let $\mathcal{N}_i$ denote its set of neighbours. The position of robot i at discrete time k is denoted by $p_i[k] \in \mathbb{R}^2$, and its dynamics are

$$p_i[k+1] = p_i[k] + u_i[k],$$

where $u_i[k] \in \mathbb{R}^2$ is the control input and it satisfies $\|u_i[k]\|_\infty \leq 0.5$. The initial positions are

$$p_1[0] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad p_2[0] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad p_3[0] = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad p_4[0] = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

So, the robots initially form a unit square. Define the desired relative displacement $\forall (i, j) \in \mathcal{E}$ as

$$d_{ij} = p_i[0] - p_j[0].$$

Use the prediction horizon $H = 6$. At time k, Robot i receives predicted trajectories $\hat{p}_j[k+\ell|k], j \in \mathcal{N}_i, \ell = 1, \ldots, H$ from its neighbours. Robot i optimizes only its own future control sequence

$$u_i[k] = \begin{bmatrix} u_i[k|k] \\ u_i[k+1|k] \\ \vdots \\ u_i[k+H-1|k] \end{bmatrix}.$$

Consider the local distributed MPC objective

$$J_i = \sum_{\ell=1}^{H} \sum_{j \in \mathcal{N}_i} q \left\| (p_i[k+\ell|k] - \hat{p}_j[k+\ell|k]) - d_{ij} \right\|_2^2 + \sum_{\ell=0}^{H-1} r \left\| u_i[k+\ell|k] \right\|_2^2,$$

where $q > 0, r > 0$. Thus, each robot attempts to preserve the initial relative formation while minimizing control effort. Define the neighbour-formation predicate

$$F_{ij} := \left\| (p_i - p_j) - d_{ij} \right\|_\infty \leq 0.2.$$

Candidate STL Specification ϕ_1:

- throughout the prediction horizon, every pair of neighbouring robots must remain close to their original relative displacement;
- sometime within the prediction horizon, all robots must reach the original square formation translated upward by 2 units.

Candidate STL Specification ϕ_2:

- throughout the prediction horizon, every pair of neighbouring robots must again remain close to their original relative displacement;
- sometime within the prediction horizon, the robots must occupy a rectangular formation whose width is larger than the original square.

Define the target positions for specification 1 and 2 respectively as

$$g_i^{(1)} = p_i[0] + \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \qquad i = 1, \ldots, 4,$$
$$g_1^{(2)} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad g_2^{(2)} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad g_3^{(2)} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad g_4^{(2)} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

The corresponding STL specification are

$$\phi_1 = \square_{[0,H]} \left(\bigwedge_{(i,j)\in\mathcal{E}} F_{ij} \right) \wedge \Diamond_{[0,H]} \left(\bigwedge_{i=1}^{4} \left\| p_i - g_i^{(1)} \right\|_\infty \leq 0.1 \right).$$
$$\phi_2 = \square_{[0,H]} \left(\bigwedge_{(i,j)\in\mathcal{E}} F_{ij} \right) \wedge \Diamond_{[0,H]} \left(\bigwedge_{i=1}^{4} \left\| p_i - g_i^{(2)} \right\|_\infty \leq 0.1 \right).$$

Q1. Formulate the local distributed MPC optimization problem solved by Robot i. Clearly identify what are the the local decision variables; the quantities known locally; and the predicted quantities received from neighbouring robots. Explain why, for fixed neighbour-predicted trajectories, the baseline local MPC problem is a convex optimization problem.

Q2. For the STL specification ϕ_1, translate the "always" formation-maintenance requirement into constraints on the predicted trajectories.

Q3. For the STL specification ϕ_2, translate the corresponding "always" formation-maintenance requirement into constraints on the predicted trajectories.

Q4. Determine which of the two STL specifications ϕ_1 or ϕ_2 results in a feasible distributed MPC problem at time $k = 0$. Your answer must be supported by a mathematical argument. A statement such as "a numerical solver reports feasible/infeasible" is not sufficient.

Q5. For the feasible STL specification, construct at least one explicit set of control sequences u_1, u_2, u_3, u_4 that demonstrates feasibility. Verify that all input constraints are satisfied; the "always" STL requirement is satisfied throughout the prediction horizon; and the "eventually" STL requirement is satisfied at some point within the prediction horizon.

Problem 3: The Chief Minister's AI Policy Simulator (50 pts)
The Hon'ble Chief Minister of Telangana, proposes to introduce a new public welfare scheme for the people of Telangana. Before launching the scheme, the Chief Minister approaches you, an AI expert, with the following question:

CM's request to you

My government is planning to introduce a new welfare scheme. I expect the scheme to increase public participation, but this may also increase the workload of government departments. Can you build an AI-based simulation of how the scheme will operate over the next 365 days, identify administrative bottlenecks, and recommend whether the scheme should be launched? Can the government departments learn to cooperate and improve their collective performance?

Your task is to construct a stochastic model of government administration and subsequently formulate the problem as a **cooperative Multi-Agent Reinforcement Learning (MARL)** problem. The objective is to evaluate whether decentralized departmental decision-making can improve government service delivery and public satisfaction.

Disclaimer

The proposed scheme, administrative workflow, and numerical parameters are hypothetical. The exercise does not attempt to reproduce the actual organizational hierarchy of the Telangana government or predict the political performance of any existing scheme.

Consider a government administration consisting of N departments, represented by the set $\mathcal{V} = \{\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_N\}$. The departments form a directed hierarchical network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{E} = \{(\mathcal{D}_i, \mathcal{D}_{i+1}) \mid i = 1, \ldots, N-1\}$. Every application submitted by a citizen must pass sequentially through all departments, starting at $\mathcal{D}_1$ and terminating at $\mathcal{D}_N$. An application can move to department $\mathcal{D}_{i+1}$ only after its processing at department $\mathcal{D}_i$ is complete. For this assignment, take $N = 5$ which are

1. $\mathcal{D}_1$: Application Intake
2. $\mathcal{D}_2$: Eligibility Verification
3. $\mathcal{D}_3$: Technical Assessment
4. $\mathcal{D}_4$: Financial Assessment
5. $\mathcal{D}_5$: Final Approval

Each department is equipped with processing officers who handle applications according to a First-In-First-Out (FIFO) discipline. The working duration is $H = 480$ minutes per day. The processing

requirements are given below.

Department	Time per application (minutes)	Initial officers
$\mathcal{D}_1$	20	3
$\mathcal{D}_2$	30	3
$\mathcal{D}_3$	60	3
$\mathcal{D}_4$	40	3
$\mathcal{D}_5$	30	3

An officer can process only one application at a time. Different officers may process different applications simultaneously. An application may pass through several departments within the same working day. If an application cannot complete all processing stages by the end of the working day, it resumes from its current processing stage on the following day. Any partially completed processing effort must be preserved; the application does not restart from $\mathcal{D}_1$. There is no overtime, and unprocessed applications remain in their departmental queues.

Let $d \in \{1, \ldots, 365\}$ denote the day index. The number of new applications received on day d is a random variable A_d satisfying $A_d \sim \text{Poisson}(\lambda_d)$, where λ_d is the expected daily demand. Initially, assume $\lambda_d = 16\theta, d = 1, \ldots, 365$. The parameter θ represents the impact of the new scheme on public applications. Consider the following four scenarios:

Scenario	θ	Interpretation
S1	0.75	Reduced demand
S2	1.00	Normal demand
S3	1.25	Increased demand
S4	1.50	High demand

a) Simulate the arrival and processing of applications over 365 days for all four scenarios.

b) Track each application individually, including its arrival day, current department, remaining processing requirement, and completion day.

c) Let $Q_i(d)$ denote the number of applications present at department $\mathcal{D}_i$ at the end of day d. Plot $Q_i(d)$ for all five departments.

d) Let B_d denote the total number of unfinished applications at the end of day d. Plot the evolution of B_d for all four scenarios.

e) Investigate whether an increase in application arrivals necessarily results in a corresponding increase in completed applications.

Assume that the citizens evaluate the government based on the timeliness of service delivery. Sup-

pose application j arrives on day a_j and completes processing on day c_j. If the application is completed on its arrival day, the government receives a reward of $+5$. If an application remains incomplete at the end of a day, the government incurs a penalty of -10 for that pending application. The daily government reward is therefore

$$r_d = 5C_d^{\text{same}} - 10B_d, \tag{1}$$

where C_d^{same} denotes the number of applications that arrive and are completed on day d. An application completed after its arrival day receives no completion bonus. Its delay penalties have already been accumulated during the days on which it remained unfinished. The cumulative government reward is

$$R_d = \sum_{k=1}^{d} r_k.$$

a) Plot R_d over 365 days for all four scenarios (S1 – S4).

b) Explain the relationship between processing bottlenecks, pending applications, and cumulative government reward.

Cooperative MARL Formulation:

The Chief Minister recognizes that the departmental staffing configuration need not remain fixed. Departments may cooperate by temporarily sharing processing officers with their immediate neighbours. Instead of prescribing a centralized staffing strategy, the government allows each department to learn its own resource-allocation policy. The five departments now become five cooperative RL agents. The set of agents is $\mathcal{N} = \{1, 2, 3, 4, 5\}$. Agent i corresponds to department $\mathcal{D}_i$. All agents cooperate to maximize the cumulative government reward. At the beginning of day d, let $Q_i^{\text{start}}(d)$ denote the number of applications present at department i, including newly arriving applications at department $\mathcal{D}_1$. Each department observes its own queue, the queues of its immediate neighbours, and the public satisfaction from the previous day. For interior departments,

$$o_i(d) = \left(Q_{i-1}^{\text{start}}(d), Q_i^{\text{start}}(d), Q_{i+1}^{\text{start}}(d), q_{d-1}\right).$$

The first and last departments observe only their existing neighbours. The quantity $q_d \in [-1, 1]$ represents public satisfaction, defined subsequently. Each agent makes its decision using its own local observation. At the beginning of every day, each agent selects an action

$$a_i(d) = \begin{cases} -1, & \text{lend one officer to } \mathcal{D}_{i-1}, \\ 0, & \text{retain all officers,} \\ +1, & \text{lend one officer to } \mathcal{D}_{i+1}. \end{cases}$$

Boundary departments cannot transfer officers to non-existent neighbours. Thus, their admissible action sets are $\mathcal{A}_1 = \{0, +1\}$, and $\mathcal{A}_5 = \{-1, 0\}$ respectively. For $i = 2, 3, 4$, $\mathcal{A}_i = \{-1, 0, +1\}$.

Assume that officers are temporarily transferable between neighbouring departments and are qualified to perform the corresponding tasks. An officer transferred to another department works there for the entire day and returns to the original department the following morning. A department may lend at most one of its original three officers per day. Borrowed officers cannot be transferred onward to another department on the same day. Let $m_i(d)$ denote the effective number of officers available at department i on day d. Since officers are only redistributed, $\sum_{i=1}^{5} m_i(d) = 15$. All five agents receive the same reward r_d given by (1). The cooperative objective is

$$\max_{\pi_1,\ldots,\pi_5} \quad \mathbb{E}\left[\sum_{d=1}^{365} \gamma^{d-1} r_d\right],$$

where π_i denotes the policy of agent i and $\gamma \in (0, 1]$ is the discount factor. The underlying environment state includes all pending applications, their processing stages, remaining service requirements, application ages, and public demand variables. Agents do not observe this complete state. They operate using their local observations.

a) Explain why this formulation is a cooperative MARL problem rather than a single-agent reinforcement learning problem.

b) Identify the agents, observations, actions, environment state, transition mechanism, and shared reward.

c) Consider a situation where $\mathcal{D}_3$ is heavily congested while $\mathcal{D}_2$ has very few pending applications. Explain how officer sharing could improve the overall processing performance.

d) Can transferring an officer from one department to another ever worsen the global reward? Justify your answer.

Let S_d denote the number of applications received on day d that are successfully completed on the same day. Define the on-time completion fraction as

$$p_d = \frac{S_d}{A_d}, \qquad A_d > 0.$$

If $A_d = 0$, use $p_d = p_{d-1}$, with $p_0 = 0.5$. Public satisfaction on day d, denoted by q_d evolves according to

$$q_d = 0.8q_{d-1} + 0.2(2p_d - 1), \quad \text{with } q_0 = 0.$$

The expected number of applications on the following day and the actual arrivals are generated respectively by

$$\lambda_{d+1} = 16\theta(1 + 0.5q_d).$$
$$A_{d+1} \sim \text{Poisson}(\lambda_{d+1}).$$

Therefore, better service improves public satisfaction and potentially increases future demand. Conversely, delayed service reduces public satisfaction and discourages future applications.

a) Integrate the public satisfaction model into the government simulation.

b) Explain why an improvement in government service delivery may eventually lead to increased departmental workloads.

c) Explain how the actions of a single department can indirectly influence future application arrivals across the entire government network.

d) Is it possible for a highly popular scheme to initially perform well and subsequently fail because of administrative congestion?

The Chief Minister now requests a final assessment of the proposed scheme based on the performance of the learned cooperative departmental policies. Evaluate the following administrative system where every department retains its original three officers on every day. System must use the above departmental processing requirements and public demand feedback model. Test the system under all four demand scenarios S1–S4. For every scenario and administrative system, conduct at least 20 independent 365-day evaluation simulations using different random seeds. Report the following:

a) Total number of applications received during the year.

b) Total number of completed applications.

c) Fraction of applications completed on the same day.

d) Average completion delay among completed applications.

e) Number of unfinished applications at the end of day 365.

f) Cumulative government reward R_{365}.

g) Final public satisfaction q_{365}.

h) Average effective staffing level at each department over the year.

Based on your evaluation, should the Chief Minister launch the scheme under the existing fixed-staffing system? Your recommendation must address administrative feasibility, public satisfaction, resource allocation, and the risks associated with uncertain future demand.